

NepError

A Hierarchical Diagnostic Taxonomy of Failure Modes
in Nepali Natural Language Inference

Annotation Guidelines & Training Manual

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Before you start: Read this entire document before annotating a single row. Annotate the 50-row pilot set independently without discussing with the other annotator. Only discuss after both of you have submitted your pilot labels.

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What is This Project?

NepError is a research project studying *why* AI language models fail on Nepali text, not just *how often* they fail.

Background: What is Natural Language Inference (NLI)?

NLI is a task where an AI model reads two sentences and decides their logical relationship.

NLI: The Basic Task

Given a **Premise** (a statement assumed to be true) and a **Hypothesis** (a statement to be evaluated), classify the relationship as:

- **ENTAILMENT** — The hypothesis must be true if the premise is true.
- **NEUTRAL** — The hypothesis may or may not be true; the premise neither confirms nor denies it.
- **CONTRADICTION** — The hypothesis is false given the premise.

Example (English):

NLI Example

Premise: The child is playing football in the park.

Hypothesis: The child is outdoors.

Correct label: **ENTAILMENT** (Playing in the park means the child is outdoors.)

Premise: The child is playing football in the park.

Hypothesis: The child is wearing red shoes.

Correct label: **NEUTRAL** (Nothing tells us about the shoes.)

Premise: The child is playing football in the park.

Hypothesis: The child is sleeping.

Correct label: **CONTRADICTION** (Playing and sleeping cannot happen simultaneously.)

What Are We Annotating?

The datasets (MNLI-Nepali, RTE-Nepali, XNLI-Nepali) contain Nepali premise-hypothesis pairs. We ran four AI models on these pairs and collected all **wrong predictions** into failure files.

Your job: Read each failure case and determine *why* the model was likely wrong, using the 4-tier taxonomy defined in this document.

Important: You are *not* deciding whether the gold label is correct. You are diagnosing what *linguistic property* most likely caused the model to fail. The gold label in the CSV is the correct answer — treat it as ground truth.

The Four-Tier Taxonomy

Every failure case receives exactly **one** taxonomy code. The taxonomy has four tiers, each capturing a different type of linguistic difficulty.

Tier	Name	Codes	
T1	Surface / Input	T1a , T1b	
T2	Morpho-Syntactic	T2a , T2b	
T3	Socio-Pragmatic	T3a , T3b	
T4	Ambiguous / Noise	T4a , T4b	

T1 — Surface / Input Errors

These failures happen before the model even begins reasoning. The *input text itself* is malformed from the model’s perspective.

T1a — Tokenization Fragmentation

What it is: The model’s tokenizer splits a Nepali word at the wrong point — cutting through a meaningful suffix or morpheme. The model processes a fragment instead of the complete word, losing grammatical or semantic information.

How to detect it: Look for long or compound Devanagari words where a suffix carries important meaning (tense, negation, case). If the word is morphologically complex and the failure is inexplicable by semantics, T1a is likely. You cannot see the tokenizer output directly — you are inferring from the context.

T1a

Premise: *Ramesh bazar gaye.*

रमेश बजार गए। (Ramesh went to the market.)

Hypothesis: *Ramesh ghar gaye.*

रमेश घर गए। (Ramesh went home.)

Gold label: **CONTRADICTION** **Model prediction:** **ENTAILMENT**

Diagnosis: Both sentences share the verb *gaye* (went). A tokenizer that fragments the spatial noun *bazar* (market) may strip the directional context, causing the model to see two near-identical sentences and incorrectly predict entailment.

Assign: **T1a**

T1b — Romanization / Script Noise

What it is: The sentence is written in the Roman alphabet instead of Devanagari script, or contains heavily transliterated Nepali. Models trained on Devanagari treat this as out-of-vocabulary (OOV) noise.

How to detect it: The premise or hypothesis is written in Latin script (a, b, c...) rather than Devanagari. This is immediately visible.

T1b

Premise (Roman): “manche bihana uthyo”

Hypothesis (Roman): “manche sutyo”

Gold label: **CONTRADICTION** **Model prediction:** **NEUTRAL**

Diagnosis: Entire input is in Roman script. The model has no Devanagari to process and treats the tokens as unknown words.

Assign: **T1b****T2 — Morpho-Syntactic Errors**

These failures come from the model misreading grammatical structure — specifically, morphological markers (suffixes, particles) that change meaning fundamentally.

T2a — Negation Particle Drop

What it is: Nepali encodes negation through verb suffixes and particles. The model ignores a negation marker that *completely inverts* the logical relationship.

Negation markers to watch for:

Devanagari	Romanization	Meaning
छैन	chhaina	is not / does not
छैनन्	chhainan	are not (plural/hon.)
दैन	daina	does not (habitual)
दैनन्	dainan	do not (plural)
होइन	hoina	is not / no
होइनन्	hoinan	are not
नाई	nai	not (emphatic)
नगर्	nagar	do not (imperative)
नहुने	nahune	should not happen
नभएको	nabhaeko	not having been
नगरेको	nagareko	not having done

How to detect it: Look for any of the above markers. Then ask: *if the negation were removed, would the model's wrong prediction become correct?* If yes, this is T2a.

T2a

Premise: रमेशले खाना खाएन।

Ramesh le khaana khayena. (Ramesh did not eat food.)

Hypothesis: रमेश भोको छ।

Ramesh bhoko chha. (Ramesh is hungry.)

Gold label: **NEUTRAL** **Model prediction:** **CONTRADICTION**

flag_negation = 1

Diagnosis: The negation suffix *-ena* on *khayena* (did not eat) is present. Remove the negation: “Ramesh ate food” + “Ramesh is hungry” would be **NEUTRAL** (eating doesn't confirm hunger). But the model predicted **CONTRADICTION**, likely because it read “ate food” as incompatible with “hungry”. The negation confused the direction of inference.

Assign: **T2a**

Critical: The CSV column `flag_negation = 1` tells you a negation word is *present*. It does NOT confirm the marker caused the failure. You must verify causality: does removing the negation explain the wrong prediction? If yes: **T2a**. If the failure seems caused by something else: continue the checklist.

T2b — Honorific / Verb Agreement Clash

What it is: Nepali has a three-level honorific system encoded in verb endings. The model fails to track which register a speaker is using, misidentifying the subject or the relationship between actors.

The three registers:

- **Low** (तँ / ta) — used for children, very close friends, inferiors
- **Middle** (तिमी / timi) — neutral / familiar
- **High** (हजुर / hajur) — formal, respectful; triggers different verb endings

T2b

Premise: हजुर कहाँ जानुहुन्छ?

Hajur kaha jaanu huncha? (Formal: Where are you [hon.] going?)

Hypothesis: त्यो मान्छे बजार जान्छ।

Tyo manchhe bazar jancha. (That person goes to the market.)

Gold label: NEUTRAL **Model prediction:** ENTAILMENT

Diagnosis: The model failed to separate the honorific reference *Hajur* (the person being addressed formally) from the third-person neutral *tyo manchhe* (that person). It treated both as the same entity and incorrectly inferred entailment.

Assign: T2b

T3 — Socio-Pragmatic Errors

These failures arise from the social and cultural context of language use.

T3a — Code-Switching (Neplish)

What it is: The sentence mixes Devanagari Nepali and English words within the same clause or sentence. This “Neplish” style is common in urban Kathmandu speech. Models struggle because consecutive tokens belong to different linguistic systems.

How to detect it: English words are embedded within Devanagari text in the *same sentence*. Note: the CSV column `flag_codeswitching` may show 0 in machine-translated data even when mixing is present — always verify visually.

T3a

Premise (Neplish): *Usle office ma presentation diyeko thiyo, tara boss le approve garena.*
(She had given a presentation at the office, but the boss did not approve it.)

Hypothesis: *Presentation successful bhayo.*
(The presentation was successful.)

Gold label: CONTRADICTION **Model prediction:** NEUTRAL

Diagnosis: The words “office”, “presentation”, “boss”, “approve” are English embedded in Devanagari structure. The model processed the mixed tokens poorly and failed to identify the explicit negation *garena* (did not approve) that makes the hypothesis a contradiction.

Assign: T3a

Note: In MNLI-Nepali and RTE-Nepali (machine-translated), T3a cases are rare. This code will be heavily used for the 150 natively authored pairs.

T3b — Domain Vocabulary Gap

What it is: The sentence contains technical vocabulary from a domain the model rarely encountered during training. Key domains for Nepali NLP: **legal/judicial**, **rural agriculture**, **community forestry**, **medical/clinical**.

How to detect it: Identify words that are domain-specific and would be rare in general Nepali text. If the failure seems to hinge on one or two specific technical words, suspect T3b.

T3b

Premise: जङ्गल पञ्चायतको वन उपभोक्ताको नियमानुसार काठ काट्न मना गरेको छ।
(The Jungle Panchayat has prohibited wood-cutting per community forest user rules.)

Hypothesis: वनमा काठ काट्न पाउँछ।
(It is permissible to cut wood in the forest.)

Gold label: **CONTRADICTION** **Model prediction:** **NEUTRAL**

Diagnosis: जङ्गल पञ्चायत (Jungle Panchayat) and वन उपभोक्ता (forest user group) are community forestry governance terms. Without knowing these terms, the model cannot parse the prohibitive statement and defaults to neutral.

Assign: **T3b**

T4 — Ambiguous / Noise

These are cases where the failure originates in the *quality or inherent difficulty of the data itself*, not in the model’s architecture.

T4a — Genuine Semantic Ambiguity

What it is: The pair is genuinely ambiguous. Even two careful native Nepali speakers, reading independently, would assign different labels. The model’s “failure” may actually be a defensible reading.

How to detect it: Ask: *Is there a plausible interpretation under which the model’s prediction is reasonable?* If yes, and you cannot definitively say the gold label is the only correct answer, this is T4a.

T4a

Premise: रमेश र सिता एकलै काम गर्ने।
Ramesh ra Sita ekle kaam garne. (Ramesh and Sita each work alone.)

Hypothesis: तिनीहरू सँगै काम गर्दैनन्।
Tini haru sangai kaam gardainan. (They do not work together.)

Gold label: **ENTAILMENT** **Model prediction:** **NEUTRAL**

Diagnosis: *Ekle kaam garne* can mean “each works independently” but could also mean they work on their own projects without implying they never collaborate. A native speaker could reasonably label this **NEUTRAL** or **ENTAILMENT**. The model’s neutral prediction is defensible.

Assign: **T4a**

T4b — Translation Artifact

What it is: MNLI-Nepali and RTE-Nepali were created by machine-translating English using GPT-4o-mini and Gemini. Some sentences are unnatural, grammatically broken, or semantically distorted. When the *Nepali text itself is broken*, the model's failure is caused by the data, not a linguistic phenomenon.

How to detect it: Read the sentence. Does it sound like natural Nepali? Does the meaning feel coherent? If the sentence is garbled, has wrong word order, or would never be written by a native speaker — this is T4b.

T4b

Premise: *Manisharuko samuhik sangathan milera samajik karya-byapar garne garchha.*

(Rough back-translation: People-of collective organization together social activities business doing does.)

Hypothesis: *Yiniharu ekantama kaam garchha.*

(They work in isolation.)

Gold label: **CONTRADICTION** **Model prediction:** **NEUTRAL**

Diagnosis: The premise is syntactically unnatural. No native Nepali speaker would stack nominalized clauses this way. The machine translation failed structurally. The model's failure is caused by the unreadable premise.

Assign: **T4b**

The Decision Checklist**Annotator Decision Checklist — Apply Strictly in Order**

For each failure case, go through these questions in sequence. **Stop at the first YES** and assign that code.

Step 1. Is the premise or hypothesis written predominantly in Roman/Latin script?

YES → **T1b**

Step 2. Is Devanagari Nepali mixed with English words in the *same sentence*?

YES → **T3a**

Step 3. Does the sentence contain a morphologically complex Devanagari word where the model appears to have ignored a suffix, and no other explanation fits better?

YES → **T1a**

Step 4. Is a negation marker present (छैन, होइन, दैन, नाई, etc.) **AND** does removing the negation explain the model's wrong prediction?

YES → **T2a**

Step 5. Are honorific pronouns/verb registers present (हजुर, तिम्री, तँ) **AND** does the model appear confused about who is being referred to?

YES → **T2b**

Step 6. Does the failure hinge on a rare technical word from legal, agricultural, forestry, or medical domains?

YES → **T3b**

Step 7. Would two careful native Nepali speakers, reading independently, likely disagree on the correct label?

YES → **T4a**

Step 8. Does the Nepali text sound unnatural, garbled, or machine-translated?

YES → **T4b**

If none apply: Write a description in `annotator_note` and flag for discussion with the project lead.

One code per case. Every failure gets exactly one taxonomy code. If multiple conditions seem true, the checklist order determines the code — the first YES wins. Do not write compound codes like “T2a+T4b”.

How to Fill the CSV

The failure CSV files have these columns. You fill in the last three.

Column	Description
<code>instance_id</code>	Unique row ID — do not edit
<code>premise</code>	The premise sentence in Nepali
<code>hypothesis</code>	The hypothesis sentence in Nepali
<code>true_label</code>	Correct gold label (0=entailment, 1=neutral, 2=contradiction)
<code>pred_label</code>	The model’s wrong prediction
<code>correct</code>	Always False in failure files
<code>model</code>	Which model made this prediction
<code>dataset</code>	Which dataset this came from
<code>conf_label0/1/2</code>	Model confidence scores (XLM-R and mDeBERTa only)
<code>flag_negation</code>	1 if a negation marker is present in the text
<code>flag_codeswitching</code>	1 if code-switching was detected
<code>flag_romanization</code>	1 if Roman script was detected
<code>taxonomy_tier</code>	YOU FILL: one of T1a, T1b, T2a, T2b, T3a, T3b, T4a, T4b
<code>annotator_note</code>	YOU FILL: brief justification (1–2 sentences)
<code>annotator_id</code>	YOU FILL: your assigned annotator code

Label Encoding Reference

Number	MNLI / XNLI	RTE
0	ENTAILMENT	ENTAILMENT
1	NEUTRAL	not_entailment
2	CONTRADICTION	(not used in RTE)

Annotator Note Format

Note Format Examples

Good (T2a): “Negation marker *chhaina* present; removing it would make model’s entailment prediction correct — negation caused the failure.”

Good (T4b): “Premise reads as broken machine translation; stacked nominalization makes core meaning unclear. Model defaulted to neutral.”

Good (T4a): “Both neutral and entailment are defensible interpretations; premise is underspecified about the nature of the relationship.”

Bad: “The model was wrong.” (Too vague — provides no evidence for the paper.)

Worked Examples — Full Walkthroughs

These five examples walk through the full decision process. Read them carefully before starting annotation.

Example 1 — T2a (Negation Drop)

Premise: उसले पैसा तिरेन।

Usle paisa tirena. (He did not pay the money. “-ena” is the past-tense negation suffix.)

Hypothesis: उसले ऋण चुकायो।

Usle rin chukaayo. (He repaid the debt.)

Gold label: **CONTRADICTION** **Model prediction:** **ENTAILMENT**
flag_negation = 1

Decision process:

1. Roman script? No.
2. Code-switching? No.
3. Tokenization fragmentation? Not obvious.
4. Negation marker? **Yes** — *tirena* contains “-ena” (did not pay). Does removing it explain the wrong prediction? Without negation: “He paid the money” + “He repaid the debt” = **ENTAILMENT**. Yes. **STOP**.

Code: **T2a** **Note:** “Negation suffix -ena on ‘tirena’ (did not pay) inverts the label to CON; model likely read ‘tire’ (paid) and inferred entailment from debt repayment.”

Example 2 — T4b (Translation Artifact)

Premise: *Sabai byaktiharuko manchheko adhikarko vyavastha sambandhi samajik nyayako paramparagat siddhantaharulai prabhabkari rupama laagu garnu parne chha.*

(Back-translation: All persons-of-human-rights-provision-related-social-justice-traditional-principles must-be-implemented effectively.)

Hypothesis: *Manchheko adhikarko karyaannvayan garnu parchha.*

(Human rights must be implemented.)

Gold label: **ENTAILMENT** **Model prediction:** **NEUTRAL**

Decision process:

1. Roman script? No.
2. Code-switching? No.
3. Tokenization? Possible, but not the dominant issue.
4. Negation? No negation markers.
5. Honorific clash? No.
6. Domain vocabulary? Legal terms present, but the sentence is also structurally broken.
7. Genuine ambiguity? No — if understood, entailment is clear.
8. Translation artifact? **Yes**. The premise is an incomprehensible stack of compound nouns. No native Nepali speaker writes this way. Machine translation failed structurally. **STOP**.

Code: **T4b** **Note:** “Premise is a syntactically malformed machine-translation artifact; multi-level compound nominalization makes the sentence unreadable even for a native speaker.”

Example 3 — T3b (Domain Vocabulary)**Premise:** भूमि सुधार आयोगको सिफारिसमा राइजत जग्गा खरिदार नियुक्त गरियो।*Bhumi sudhar aayogko sifarisma raijat jagga kharidar niyukt gariyo.*

(On the recommendation of the Land Reform Commission, a land revenue clerk was appointed.)

Hypothesis: खरिदार सरकारी कर्मचारी हो।*Kharidar sarkaari karmachari ho.* (The Kharidar is a government employee.)**Gold label:** ENTAILMENT **Model prediction:** NEUTRAL**Decision process:**

1. Roman script? No.
2. Code-switching? No.
3. Tokenization? No obvious issue.
4. Negation? No.
5. Honorific clash? No.
6. Domain vocabulary? **Yes.** *Bhumi sudhar aayog* (Land Reform Commission), *rai jat jagga* (government land), and *Kharidar* (a specific civil service position) are all legal-administrative terms. Without knowing that a Kharidar is a government-appointed officer, the model cannot make the entailment inference. **STOP.**

Code: T3b **Note:** “‘Kharidar’ is a government civil service position; model lacking this domain knowledge cannot infer entailment from ‘appointed by commission’ to ‘government employee.’”

Example 4 — T4a (Genuine Ambiguity)**Premise:** उसले धेरै मेहनत गर्छ।*Usle dherai mehnat garchha.* (He works very hard.)**Hypothesis:** उसको काम सफा छ।*Usko kaam safa chha.* (His work is clean / neat.)**Gold label:** NEUTRAL **Model prediction:** ENTAILMENT**Decision process:**

1. Roman script? No.
2. Code-switching? No.
3. Tokenization? No.
4. Negation? No.
5. Honorific? No.
6. Domain vocabulary? No.
7. Genuine ambiguity? **Yes.** In Nepali cultural context, a hardworking person (*mehnat garne manchhe*) is commonly associated with neat, quality output. Both **NEUTRAL** (no strict logical link) and **ENTAILMENT** (cultural inference) are defensible readings. **STOP.**

Code: T4a **Note:** “Cultural inference ‘hard work implies neat output’ is plausible in Nepali context; entailment is a defensible reading despite neutral gold label.”

Example 5 — T3a (Code-Switching)

Premise (Neplish): *Usle interview ma confident dekhinchha tara job offer aayena.*
(He looked confident in the interview but the job offer did not come.)

Hypothesis: उसको जागिर लाग्यो।
Usko jagir lagyo. (He got the job.)

Gold label: **CONTRADICTION** **Model prediction:** **NEUTRAL**

Decision process:

1. Roman script dominant? No — text is Devanagari with embedded English words.
2. Code-switching? **Yes.** “interview”, “confident”, “job offer” are English words embedded in a Devanagari sentence. **STOP.**

A note on priority here: A negation (*aayena* = did not come) is also present, but Step 2 (code-switching) comes before Step 4 (negation) in the checklist. The code-switched structure is the primary reason the model missed the negation. We assign T3a, and the note explains this.

Code: **T3a** **Note:** “English tokens ‘interview’, ‘confident’, ‘job offer’ embedded in Devanagari; mixed-script processing likely prevented model from detecting negation ‘aayena’ (did not come), causing neutral instead of contradiction.”

Common Mistakes to Avoid

Mistake 1: Assigning T4b for any hard or complex sentence.

T4b is only for sentences that are *structurally broken* due to machine translation. A sentence can be logically complex and still be natural Nepali — that would be T4a, not T4b.

Mistake 2: Assigning T2a just because `flag_negation = 1`.

The flag confirms a negation word is present. You must verify it actually caused the failure. Many sentences contain negation that is irrelevant to the model’s error.

Mistake 3: Assigning two codes.

One code per case. Apply the checklist in order — the first YES wins.

Mistake 4: Judging the gold label.

You are not here to correct the dataset. If the gold label says **CONTRADICTION** and the model predicted **NEUTRAL**, explain why the model went wrong — not whether the dataset is right.

Mistake 5: Leaving `annotator_note` blank.

Every row needs a note. Even one sentence. The note is what makes the annotation usable for the paper.

The Pilot Protocol

1. Both annotators independently annotate the **same 50 rows** from `failures_xlmr_mnli.csv`.
2. No communication during the pilot. Submit your labels before any discussion.
3. The project lead computes Cohen’s Kappa (κ) using `sklearn.metrics.cohen_kappa_score`.
4. **Target:** $\kappa \geq 0.70$ (substantial agreement, Landis & Koch 1977).

- If $\kappa < 0.70$: calibration session to identify confused tier pairs, then re-pilot on a fresh 50 rows.

Interpreting Cohen's Kappa

κ Range	Interpretation
< 0.20	Slight agreement
0.21–0.40	Fair agreement
0.41–0.60	Moderate agreement
0.61–0.80	Substantial agreement (our minimum target)
0.81–1.00	Almost perfect agreement

Annotation Priority Order

Within each file, annotate rows in this order:

- First — the 18 cross-model overlap instances.** These are the instances all three encoder models (XLM-R, mDeBERTa, CHiPSAL-BERT) failed on simultaneously. They are flagged in the cross-model comparison file. These are the most diagnostically valuable cases.
- Second — all `flag_negation = 1` rows.** Approximately 140 per 600-row file. These are priority T2a candidates.
- Third — random sample** from remaining rows to reach the 150–200 target.

Quick Reference Card

Code	Name	One-line test	Step
T1a	Tokenization Frag.	Suffix chopped; failure inexplicable otherwise	3
T1b	Romanization Noise	Sentence is in Latin/Roman script	1
T2a	Negation Drop	छैन / होइन / दैन present AND explains wrong label	4
T2b	Honorific Mismatch	हजुर / तिमि confusion causes reference error	5
T3a	Code-Switching	English words inside Devanagari sentence	2
T3b	Domain Vocabulary	Legal / agri / forestry / medical rare term	6
T4a	Genuine Ambiguity	Two native speakers would disagree on label	7
T4b	Translation Artifact	Sentence reads as broken / unnatural Nepali	8

Questions? Contact Pema Tshering Sherpa at ptssherpa5@gmail.com